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Synergistic association between cytochrome *bd*-encoded *Proteiniphilum* and reactive oxygen species (ROS)-scavenging methanogens in microaerobic-anaerobic digestion of lignocellulosic biomass

Zhuoying Wu[a,1], Duc Nguyen[b,1], Theo Y.C. Lam[a,c], Huichuan Zhuang[c], Shilva Shrestha[d], Lutgarde Raskin[d], Samir Kumar Khanal[b,*], Po-Heng Lee[a,*]

[a] Department of Civil and Environmental Engineering, Imperial College, London, The United Kingdom
[b] Department of Molecular Biosciences and Bioengineering, University of Hawai'i at Mānoa, Honolulu, HI 96822, USA
[c] Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong SAR
[d] Department of Civil and Environmental Engineering, University of Michigan, 1351 Beal Avenue, 107 EWRE Building, Ann Arbor, MI 48109-2125, USA

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ABSTRACT

Intermittent (every other day) microaerobic [picomolar oxygen by oxidation-reduction potential (ORP) set at +25 mV above anaerobic baseline] digestion of lignocellulosic biomass showed higher digestibility and better stability at a high organic loading rate (OLR) of 5 g volatile solids (VS)/L/d than that under strict anaerobic conditions. However, the microbial mechanisms supporting the delicate balance under microaeration remain underexplored. On the basis of our previous findings that microbial communities in replicate experiments were dominated by strains of the genus *Proteiniphilum* but contained diverse taxa of methanogenic archaea, here we recovered related genomes and reconstructed the putative metabolic pathways using a genome-centric metagenomic approach. The highly enriched *Proteiniphilum* strains were identified as efficient cellulolytic facultative bacterium, which directly degraded lignocellulose to carbon dioxide, formate, and acetate via aerobic respiration and anaerobic fermentation, alternatively. Moreover, high oxygen affinity cytochromes, *bd*-type terminal oxidases, in *Proteiniphilum* strains were found to be closely associated with such picomolar oxygen conditions, which has long been overlooked in anaerobic digestion. Furthermore, hydrogenotrophic methanogenesis was the prevalent pathway for methane production while *Methanosarcina, Methanobrevibacter*, and *Methanocorpusculum* were the dominant methanogens in the replicate experiments. Importantly, the two functional groups, namely cellulolytic facultative *Proteiniphilum* strains and methanogens, encoded various antioxidant enzymes. Energy-dependent reactive oxygen species (ROS) scavengers (superoxide reductase (*SOR*) and rubrerythrin (*rbr*) were ubiquitously present in different methanogenic taxa in response to replicate-specific ORP levels (-470, -450 and -475 mV). Collectively, cytochrome *bd* oxidase and ROS defenders may play roles in improving the digestibility and stability observed in intermittent microaerobic digestion.

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1. Introduction

In recent years, anaerobic digestion (AD) of lignocellulosic biomass has drawn significant interest for sustainable bioenergy production (Sawatdeenarunat et al., 2015). However, the highly recalcitrant nature coupled with the heterogeneous structure of lignocellulosic feedstocks impedes their digestibility. Moreover, long-term AD operation with such feedstocks at high organic loading rates (OLRs) usually leads to volatile fatty acids (VFAs) build-up, subsequent pH drop, and eventual system failure (Nguyen et al., 2019; Shrestha et al., 2017). Recently, these issues have been overcome by using a microaeration approach, a highly efficient and economically feasible strategy to improve digester performance (Botheju and Bakke, 2011; Charles et al., 2009; Fdz-Polanco et al., 2009; Johansen and Bakke, 2006; Lim and Wang, 2013; Nguyen and Khanal, 2018; Nguyen et al., 2019). Specifically, an oxidation-reduction potential (ORP)-based oxygenation approach, where the ORP level varies linearly with the logarithmic dissolved oxygen (DO) concentration, allows for a more precise control of oxygen/air injection (far below 0.1 mg/L) into a digester. ORP-based microaeration has been examined in

* Corresponding authors:
E-mail addresses: khanal@hawaii.edu (S.K. Khanal), po-heng.lee@imperial.ac.uk (P.-H. Lee).
[1] Note: Zhuoying Wu and Duc Nguyen equally contributed to this study.

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fermentation industries for ethanol and butanol production (Liu et al., 2013). Its application in AD system, however, has been very limited (Khanal and Huang, 2003; Nguyen et al., 2019).

In general, successful biogas production in AD systems depends on an intricate balance between multiple functional microbial groups, including hydrolytic bacteria, fermentative bacteria, syntrophic acetogenic bacteria, and methanogenic archaea (Briones and Raskin, 2003). Nevertheless, injection of oxygen, a redox-active gas, into AD system has been reported to stimulate the extracellular ORP level, thereby altering the intracellular redox balance, and redirecting the metabolic pathways (Banerjee, 2012), consequently reshaping the microbial community to achieve a new balance. Previous studies reported that microaeration led to a high abundance of *Firmicutes* (Fu et al., 2016; Lim et al., 2014; Tang et al., 2004; Yin et al., 2016; Nguyen et al., 2019). Since bacteria whining the *Firmicutes* are known to produce extracellular enzymes such as cellulases, lipases, and proteases (Leven et al., 2007), their high abundance is likely responsible for enhanced hydrolysis. Moreover, archaeal population shifts were also observed and were attributed to species-specific differences in oxygen tolerance (Fu et al., 2016; Tang et al., 2004). When introducing O_2, various reactive oxygen species (ROS) (e.g., superoxide radical ($O_2^{\cdot-}$), hydrogen peroxide (H_2O_2), and hydroxyl radical ($^{\cdot}OH$)) are readily formed, resulting in disruption of the cell membrane, proteins and DNA (Imlay, 2002). To overcome the oxidative stress, microorganisms, including methanogens, need to possess various antioxidant enzymes (e.g., superoxide dismutase (SOD), superoxide reductase (SOR), catalases (CAT), peroxidases (PXD), peroxiredoxins (PRX), and rubrerythrin (Rbr)) (Imlay, 2008). Hence, the ROS-detoxification mechanism plays a vital role for facultative and obligate anaerobes to adapt to the microaerobic environments in AD systems (Fischer et al., 2016). In a previous study, we developed an ORP-based intermittent microaeration AD system with picomolar O_2 addition by setting the ORP +25 mV above the strict anaerobic baseline (-470, -450, -475 mV) every other day. This system was fed with Napier grass, a typical lignocellulosic feedstock, at an OLR of 5 g VS/L/d, and achieved a stable methane yield of 80.8±9.6 N CH_4 mL/volatile solids (VS) g . Using three independent replicates, 16S rRNA and *mcrA* gene sequencing analyse showed that the microbial communities in three systems were dominated by the genus *Proteiniphilum* but contained various methanogenic taxa. This research led to several questions associated with such high-rate AD of lignocellulose under intermittent microaerobic conditions: 1) why was the same bacterial genus *Proteiniphilum* prevalent, but were diverse taxa of methanogens present in replicate systems? 2) how was each microbial population able to switch to different carbohydrate degradation pathways to adapt to intermittent microaerobic environment? and 3) how were the predominant anaerobes capable of oxygen tolerance?

To address these questions, this study was carried out to identify the functional genes involved in critical pathways within the enriched microbiomes. We used genome-centric metagenomics to gain genomic insights into the highly abundant bacterial genus *Proteiniphilum* and methanogens from two independent samples collected during ORP-based microaeration experiments. Four near-complete *Proteiniphilum* genomes and three methanogenic genomes were retrieved. Their metabolic pathways associated with lignocellulose degradation, methane formation, and oxygen detoxification were reconstructed. The functional assignments from genome sequences of these key populations were mapped to understand their physiology and behavior related to methane production. The knowledge obtained will enable biotechnological exploration of microbial interaction network under the ORP-controlled intermittent microaerobic strategy for digester design and operation.

2. Materials and Methods

2.1. Bioreactor operation and microbial sampling

As described in our previous study (Nguyen et al., 2019), a 2.0-L continuous-stirred tank reactor (CSTR) equipped with an ORP-based microaeration control system was used for digestion of Napier grass, a lignocellulosic biomass. The CSTR was fed with stepwise increase in OLR from 1.5 to 5.0 g VS/L/day under anaerobic conditions, and the reactor showed sign of instability when the OLR reached 5.0 g VS/L/day. Intermittent microaeration was then initiated by injecting oxygen by controlling ORP value at +25 mV above its corresponding anaerobic baseline every other day. The experiments were replicated three times (Rep 1, Rep 2, and Rep 3) to confirm the reactor performance during anaerobic/microaerobic experimental cycles (using the same inoculum, substrate, and reactor operating conditions). However, ORP value is extremely sensitive towards the availability of electron acceptors (e.g., O_2, NAD^+) and reducing equivalents (e.g., NADH) in the system, and significantly affected by a slight variation in operating conditions such as pH, electrical conductivity, temperature, etc. Thus, different baseline ORP values of -495±7 mV, -475±15 mV and -500±20 mV were obtained in three independent replicates. To achieve microaerobic condition, the system ORP was raised by +25 mV above the baseline ORP in each replicate, resulting in different ORP values (-470 mV, -450 mV and -475 mV in Rep1, Rep2 and Rep3, respectively). While reactor performance and microbial community (16S rRNA gene-based bacterial analysis, and *mcrA* gene-based methanogen analysis) under each scenario were examined in our previous study (Nguyen et al., 2019), the current study evaluated the microaeration-driven digestion pathway via metagenomic analysis. Here, each sample was collected when its steady-state microaeration performance was achieved [no significant difference (ANOVA, $p > 0.05$) in each operating parameter during five consecutive days of operation] in each replicate. All collected samples were centrifuged at 10,000 g for 2 min, and the pellets were stored at -80 °C until further processing. In this study, only samples in Rep 1 and Rep 2 met the sequence criteria for metagenomic analysis, whereas samples in Rep 3 were discarded due to severe DNA degradation. The reference genome sequences based on community analysis data of Rep 3, however, were chosen for comparison in this study as well.

2.2. DNA extraction and sequencing

Genomic DNA was extracted using the PowerSoil DNA Isolation Kit (Mo Bio Laboratories, Carlsbad, CA, USA) according to the manufacturers' protocols. The quality and the quantity of the extracted DNA were determined using NanoDrop (ThermoFisher Scientific, Waltham, MA, USA) and Qubit fluorometer (Life Technologies, Carlsbad, CA, USA). Genomic DNA was sent to BGI (Shenzhen, China) for paired-end library construction. The prepared libraries were sequenced on an Illumina HiSeq 2000 platform (Illumina, San Diego, CA, USA) to generate 125 bp paired-end sequencing reads.

2.3. Metagenomic assembly and binning

Raw paired-end reads from the Illumina HiSeq 2000 platform were initially filtered using Trimmomatic (version 0.36) based on a minimum quality score of 30, a minimum sequence length of 76 bp and allowing for less than three ambiguous nucleotides (Ns) (Bolger et al., 2014). For each sample, the trimmed clean paired-end reads were further *de novo* assembled into long sequence contigs using SPAdes (version 3.9.0) based on the de Bruijn graph, with default settings (-k 19, 33, 47, 61, 75) (Bankevich et al., 2012;

Nurk et al., 2017). The quality of the assembled contigs was evaluated with Quast (version 4.5) (Gurevich et al., 2013). Then, metagenomic contigs were clustered into genomic bins with MaxBin based on an expectation-maximization algorithm (Wu et al., 2014). CheckM (version 1.0.18) was introduced to estimate the completeness and contamination of each draft genome using a broader set of marker genes specific to the position of a genome within a reference genome tree and information about the collocation of these genes (Parks et al., 2015). JspeciesWS online ((http://jspecies.ribohost.com/jspeciesws) was used to calculate the average nucleotide identity (ANI) values between genomes of *Proteiniphilum* strains (Richter et al., 2016).

2.4. Gene annotation and metabolic reconstruction

Gene prediction on the contigs assigned to a genomic bin was performed using the Prodigal program (Hyatt et al., 2010)-incorporated Prokka Software (Seemann, 2014). The annotations of these genome sequences were carried out by means of KO (KEGG Orthology) assignments using BlastKOALA server at KEGG website, which characterize individual gene functions and reconstruct KEGG pathways inferring high-level functions of the organism and the ecosystem (Kanehisa et al., 2016). To predict genes encoding carbohydrate-active enzymes, the carbohydrate-active enzyme database (CAZy) annotation web- server dbCAN was used (Yin et al., 2012).

2.5. Data availability

The metagenome sequencing reads have been submitted to the NCBI's Sequence Read Archive under accession numbers SAMN13634953 to SAMN13634954. Draft genomes obtained in this study have been deposited in NCBI's Whole Genome Shotgun database under accession numbers JADLKJ000000000 to JADLKP000000000. Their functional annotations have further been deposited at the FigShare Online Database (https://figshare.com/articles/dataset/Microaerobic_digestion_genome_annotation/13270487). All other relevant data are available in the supporting information.

3. Results

3.1. Overview of metagenomic sequencing and binning

High-throughput sequencing yielded a total of 87,089,214 and 91,305,190 quality pair-end reads in Rep 1 and Rep 2, respectively (Table S1). The assembly of resulting reads generated 237,059 and 58,704 contigs with N50 of 2,427 and 3,118 bp in Rep1 and Rep2, respectively (Table S1). The assemblies (both with minimum length of 76 bp) accounted for about 96% and 99% of their respective total quality reads, suggesting their fair representativeness of microbial community in the bioreactor. Focusing on the prominent pathway during microaerobic digestion, this study only investigated the core bacterial and methanogenic populations, bacterial genus *Proteiniphilum* (repeatedly enriched with abundance of 31.8, 64.5 and 43.2 %, respectively, in Rep1, Rep 2 and Rep3) and various methanogens (*Methanosarcina* (56.8 % in Rep1), *Methanobrevibacter* (53.9 % in Rep2), and *Methanocorpusculum* (45.9 % in Rep3)), as elucidated in our previous community analysis (Nguyen et al., 2019). As a result of metagenomic binning, Rep1 yielded two *Proteiniphilum* genomes (Rep1_Bin001, Bin026) and two methanogenic genomes (Rep1_Bin027, Bin035), yet without genome of *Methanosarcina* (Table 1). Rep2 resulted in two genomes of *Proteiniphilum* (Rep2_Bin001, Bin003) and one genome of *Methanobrevibacter* (Rep2_Bin008) (Table 1). For comparison and curation, representative genomes of genera *Methanosarcina, Methanobrevibacter, Methanocorpusculum*, as well as *Proteiniphilum*, were chosen as references in our study. ANI analysis (Table S2) showed that Rep1_Bin001 and Rep2_Bin001 shared a genome similarity of 97.7%, implying that highly reproducibility of bacterial population upon microaeration (at ORP of+25 mV above the anaerobic baseline). In addition, genomic mapping (Table S3) showed that the relative abundances of 43.4% of reads in Rep1 and 84.5% of reads in Rep2 were mapped to Rep1_Bin001 and Rep2_Bin001, respectively, indicating their functional importance during microaeration.

3.2. Cellulose and xylan hydrolysis by Proteiniphilum strains

The observed enhanced degradation of lignocellulosic biomass in ORP-based microaeration AD system, suggests that the highly abundant bacterial genus *Proteiniphilum* is potentially responsible for polysaccharides hydrolysis, especially cellulose and xylan. Four genomes belonging to this genus were retrieved and screened for the presence of carbohydrate-active enzymes by applying the CAZy annotation server dbCAN. For comparative purposes, the genome of a phylogenetically closely related reference strain *Proteiniphilum saccharofermentans* M3/6 (LT605205) was also investigated. Results presented in Figs. 1 and 2, and Table S4 indicate that all analyzed genomes harbor a diverse repertoire of enzymes involved in stepwise hydrolysis, including glycoside hydrolases (GH), carbohydrate esterase (CE), and substrate-specific carbohydrate-binding motifs (CBM). Among them, strain Rep2_Bin003 comprised 239 such enzymes, the highest among all four strains, followed by strain Rep2_Bin001 and strain Rep1_Bin001 with 183 and 190 related enzymes, respectively, and strain Rep1_Bin026 containing the lowest with 153 enzymes.

The complete hydrolysis of cellulose to glucose requires a combination of several enzymes working synergistically. Endoglucanases and exoglucanases (cellobiohydrolases) initially cleave the β-1,4 glycosidic bonds of long-chain cellulose, and then β-glucosidases (BGL) and cellobiose phosphorylases further break cellobioses into glucose monomers (Bhat and Bhat, 1997; Doi and Kosugi, 2004). In this study, the genomes of strains Rep1_Bin001, Rep2_Bin001, and Rep2_Bin003, encoded GH5, GH9 endoglucanases, and GH3 β-glucosidases (*bglX*), indicating their ability to hydrolyze cellulose completely similar to the reference strain M3/6. Besides, these three genomic bins contained an additional GH94 cellobiose phosphorylase, allowing versatile pathways in cellulose degradation. In addition, CBM4 and CBM6, known to bind to cellulose extracellularly (Gilbert et al., 2013), were also found in these three genomes, but were not found in the reference strain M3/6. This suggests that the *Proteiniphilum* strains enriched under intermittent microaerobic condition provided more efficient hydrolysis than that of strain M3/6. However, strain Rep1_Bin026 lacked the enzyme cellulase.

Xylan as the major polysaccharide of hemicellulose requires a synergistic action of endo-1,4-β-xylanase (XLN) and 1,4-β-xylosidase (BXL) cleaving the backbone for hydrolysis, and the accessory enzymes cleaving the side chains of the associated oligo-, di-, monosaccharides, including α-L-arabinofuranosidase, α-glucuronidase, acetyl xylan esterase and feruloyl esterase (Uffen, 1997). In this study, we found that strains Rep1_Bin001, Rep2_Bin001, and Rep2_Bin003 possessed the same enzymes responsible for xylan degradation, but different from those present in strains Rep1_Bin026 and M3/6. Strains Rep1_Bin001, Rep2_Bin001, and Rep2_Bin003 encoded GH10 xylanase (*xynA*) and GH43 xylosidase (*xynB*). Moreover, the three genomes also encoded accessory enzymes, GH43, GH51 arabinofuranosidase (*abfA*), and GH67 glucuronidase (*aguA*), for further xylan degradation into monosaccharides. In addition, the aforementioned CBM4 and CBM6, as well as CBM13 and CBM22, which were characterized by binding

Table 1
Genome statistics of four *Proteiniphilum* and three methanogenic archaeal genomes recovered from microaerobic digestion experiments

	Genomes				Contigs			Genes	Taxonomy		Abundance	Reactor
Genome Bin ID	Genome Size (bp)	GC Content (%)	Completeness (%)	Contamination (%)	No. of Contigs /	Contigs N50 (bp)	Mean contig length (bp)	No. of predicted genes /	Similarity of genomes between isolate and GenBank entry (ANI, %)	Closest related NCBI GenBank entry with a validly published taxonomic affiliation /	Mapped reads (%)	Replicates and Condition /
Rep1_ Bin001	2,516,833	49.49	96.17	0	29	137,125	86,787	2,068	71.61	*Proteiniphilum saccharofermentans* M3/6 (LT605205)	43.39	Rep1_ Micro-aeration
Rep1_ Bin026	2,846,721	51.64	97.52	4.07	300	22,581	9,489	2,484	71.92	*Proteiniphilum saccharofermentans* M3/6 (LT605205)	2.57	Rep1_ Micro-aeration
Rep1_ Bin027	2,220,121	42.41	99.20	0.00	169	28,850	13,137	2,241	98.84	*Methanobacterium formicicum* BRM9 (CP006933)	0.50	Rep1_ Micro-aeration
Rep1_ Bin035	1,867,247	33.54	78.53	0.80	464	5,108	4,024	1,661	80.24	*Methanobrevibacter olleyae* YLM1 (CP014265)	0.29	Rep1_ Micro-aeration
Rep2_ Bin001	2,788,724	48.85	100	0.55	11	477,167	253,520	2,322	71.95	*Proteiniphilum saccharofermentans* M3/6 (LT605205)	84.49	Rep2_ Micro-aeration
Rep2_ Bin003	3,328,578	51.02	96.96	5.21	319	30,718	10,434	2,926	71.87	*Proteiniphilum saccharofermentans* M3/6 (LT605205)	10.46	Rep2_ Micro-aeration
Rep2_ Bin008	2,459,061	32.84	97.60	1.20	371	9,213	6,628	2,230	80.71	*Methanobrevibacter olleyae* YLM1 (CP014265)	0.66	Rep2_ Micro-aeration



Fig. 1. Functional overview of seven studied genomic bins and their closely related reference genomes. Filled box means a complete set of genes for the metabolic pathway was found in the corresponding genome; Filled circle means only one or two genes are missing in the metabolic pathway of this genome; Empty box means the lack of several key genes in the pathway and the absence of the metabolic process in the genome. See Table S2 for a list of the key functional genes used to indicate the presence of a metabolism.

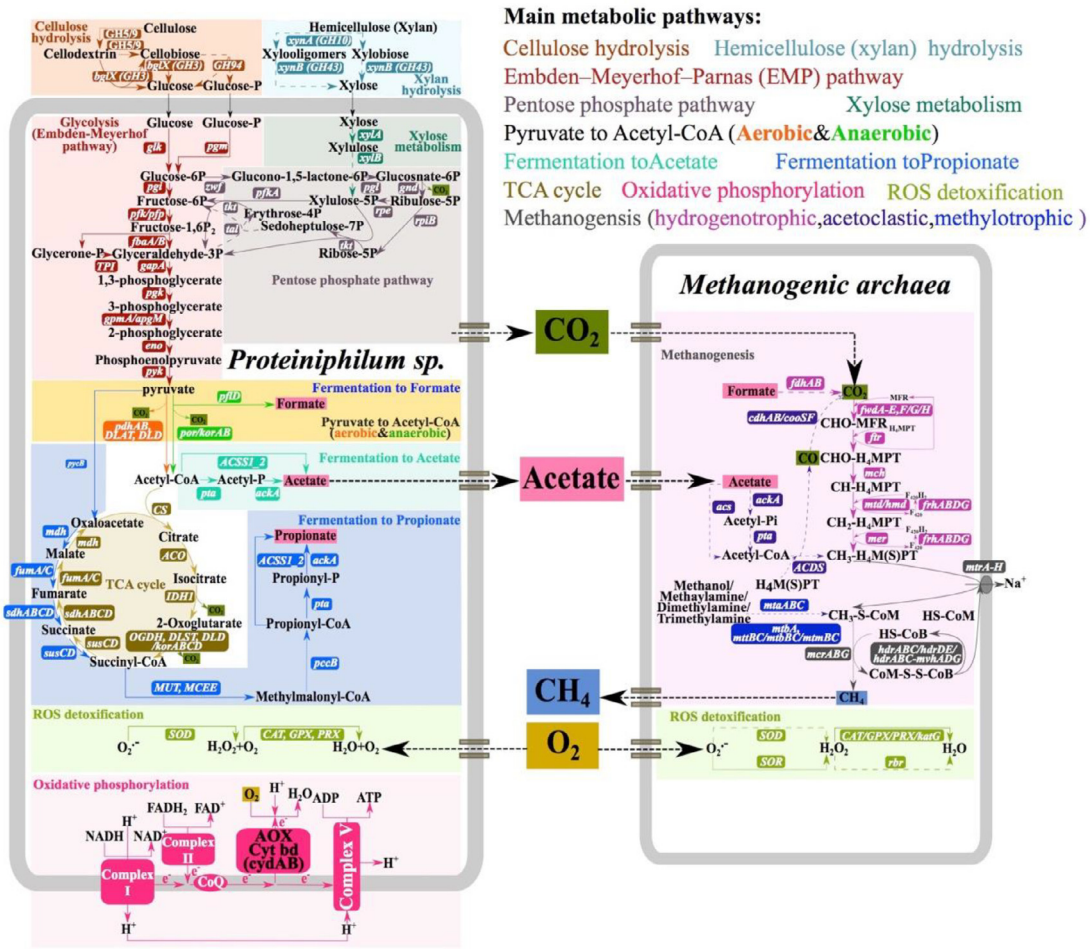


Fig. 2. Proposed metabolic pathway and interactions between *Proteiniphilum* and methanogens in intermittent microaerobic digestion process. The reconstructed pathways include cellulose hydrolysis, hemicellulose (xylan) hydrolysis, Embden–Meyerhof–Parnas (EMP) pathway, pentose phosphate pathway, xylose metabolism, pyruvate metabolism, tricarboxylic acid (TCA) cycle, oxidative phosphorylation, fermentation to formate, acetate and propionate, hydrogenotrophic, acetoclastic and methylotrophic methanogenesis, and ROS detoxification. Solid line means gene (cluster) recovered in all studied genomes; Dotted line means the gene (cluster) lacked in one of the studied genomes, but present in reference genome; And, dashed line means the gene (cluster) lacked in the studied genomes of same genus, as well as the reference genome.

specificity for linear and branched/decorated xylan (Gilbert et al., 2013), were also detected. However, strain M3/6 lacked glucuronidase, whereas strain Rep1_Bin026 lacked all xylan-related enzymes. Therefore, this analysis confirmed that *Proteiniphilum* strains capable of degrading cellulose and hemicellulose (xylan) were actively enriched in the Napier grass-fed microaerobic digesters. Meanwhile, the *Proteiniphilum* strains also showed their versatility in hydrolysis pathways.

3.3. *Carbohydrate degradation via alternative aerobic respiration and anaerobic fermentation by cytochrome* bd *encoded* Proteiniphilum *strains*

Considering the picomolar oxygen dosing every other day, it is postulated that the microorganisms may activate micro-aerobic and anaerobic metabolic switch alternatively to respond to the reducing equivalent change driven by direct oxygen variation. To further elucidate this postulation, the putative genes of the predominant *Proteiniphilum* strains were assigned to both fermentation and respiration pathways in the KEGG database (Figs. 1 and 2, and Table S4).

Given that lignocellulose hydrolysis yields hexose (e.g., glucose) and pentose (e.g., xylose) sugars, their degradation pathways were investigated. Results showed that all four *Proteiniphilum* genomes, as well as the reference strain M3/6, encoded a series of enzymes for prominent central carbon metabolic routes, i.e., the Embden-Meyerhof-Parnas (EMP), and the pentose phosphate (PP) pathways. Particularly, genomes Rep1_Bin026, Rep2_Bin001, and strain M3/6 contained a full complement of genes for EMP pathway, whereas genomes Rep1_Bin001 and Rep2_Bin003 lacked enolase (*eno*) and phosphoglycerate kinase (*pgk*), respectively. This could be attributed to the assembly gaps. Moreover, in contrast to strain M3/6, all four recovered *Proteiniphilum* genomes lacked transaldolase (*tal*) involved in the non-oxidative PP pathway. Instead, they might perform an alternative hexose–pentose interconversion pathway in which only enzyme transketolase (*tkt*) in the PP pathway would be involved. Xylulose was catalyzed from xylose, a pentose from xylan hydrolysis, via the action of xylose isomerase (*xylA*) and xylulose kinase (*xylB*), as observed in all these genomes. Thus, the *Proteiniphilum* strains in this study may mainly use the EMP pathway for hexose degradation, or alternatively use the incomplete PP pathway integrated with the EMP pathway for pentose degradation.

Pyruvate, the end-product of EMP pathway, serves as a crucial intermediate in the central metabolic network, linking fermentation or respiration for a better adaptation in different oxygen-mediated redox environments. Genome analysis revealed that all four analyzed *Proteiniphilum* genomes, as well as the reference strain M3/6, harbored a number of enzymes for mixed acid fermentation from pyruvate. They encoded pyruvate formate

lyase (*pflD*), allowing formate and acetyl-CoA formation via non-oxidative conversion. Meanwhile, they also contained several ferredoxin/flavodoxin oxidoreductases (PFOR, gene *por* or *korAB*) encoding for oxidative decarboxylation of pyruvate to acetyl-CoA anaerobically. Subsequently, the aforementioned acetyl-CoA is capable of further converting into acetate via either a single-step reaction by an acetyl-CoA synthetase (*acs*) or a two-step reaction by phosphate acetyltransferase (*pta*) and acetate kinase (*ackA*) in all the four *Proteiniphilum* genomes. However, all retrieved *Proteiniphilum* strains lacked methylmalonyl-CoA carboxytransferase (*pccB*) enzyme in methylmalonyl-CoA pathway for propionate production, while the strain M3/6 possessed this enzyme. Furthermore, the common pathway for production of intermediates such as ethanol, butyrate and lactate, were incomplete or missing in these genomes. As a result, it is believed that the *Proteiniphilum* strains studied can form formate and acetate via anaerobic fermentation. Although strain M3/6 have been verified to produce acetate and propionate (Hahnke et al., 2016), fermentative propionate production in the *Proteiniphilum* strains present in our experiments should be further examined by physiological studies.

In addition to anaerobic fermentation, the *Proteiniphilum* strains also encoded all major components involved in aerobic respiration, including pyruvate dehydrogenase complex (PDHC, gene *pdhAB, DLAT* and *DLD*), tricarboxylic acid (TCA) cycle, and electron transfer complex. The pyruvate PDHC acted as a substitution of PFOR for pyruvate decarboxylation to acetyl-CoA but functioned under aerobic conditions (Ginger et al., 2010). Then, TCA cycle would further oxidize acetyl-CoA completely into CO_2, with addition of electron carriers (NADH and $FADH_2$). Finally, these electron carriers would pass their electrons into the electron transport chain and drive ATP production through oxidative phosphorylation. Interestingly, only one type of oxygen-dependent terminal oxidases, cytochrome *bd*-quinol oxidases (*cydAB*), was identified in the *Proteiniphilum* strains, as well as in the strain M3/6. Since cytochrome *bd*-quinol oxidase has much higher affinity towards O_2 (D'mello and Poole, 1996), its presence suggests that the *Proteiniphilum* strains could respire under extremely low oxygen environments to harvest more energy compared to under anaerobic conditions, which was consistent with our experimental setup in which picomolar oxygen was injected by intermittently setting ORP value at +25 mV above anaerobic baseline.

Jointly, the genomes of *Proteiniphilum* strains contained complete sets of genes for versatile carbon and energy metabolism pathways, including anaerobic/aerobic pyruvate degradation (PFOR/PDHC) in central carbon metabolism, fermentative acetate, formate, and propionate production, TCA cycle, and electron transport-oxidative phosphorylation system using cytochrome *bd* as terminal oxidase. They provided the genetic potential for their facultative metabolism, but might not grow well under high oxygen conditions, known as "nanaerobes" (Baughn and Malamy, 2004). However, due to the favorable energy yield, such nanaerobes were likely to prefer aerobic respiration rather than anaerobic fermentation under intermittent micro-aerobic conditions.

3.4. Hydrogenotrophic methanogenesis prevalence

In our previous study, the digester recovered from the verge of collapse when ORP-controlled microaeration was introduced (Nguyen et al., 2019). Reactor performance data showed that the addition of a picomolar level of oxygen prompted the degradation of accumulated VFAs, elevated the system pH value back to neutral level, thereby increasing the methanogenic activity and subsequently the methane yield. However, contrary to consistent bacterial dominance, the dominant methanogenic populations differed among the three microaeration replicates (*Methanosarcina* in Rep1, *Methanobrevibacter* in Rep2, and *Methanocorpusculum* in Rep3) based on *mcrA* gene analysis (Nguyen et al., 2019). In order to investigate the major methanogenesis route, this study retrieved methanogenic genomes (Figs. 1 and 2, and Table S4). Rep1 yielded two genomes related to genera *Methanobacterium* (Rep1_Bin027) and *Methanobrevibacter* (Rep1_Bin035), without dominant *Methanosarcina*, but did not recover a genome consistent with the dominant *Methanosarcina*. Rep2 produced one genome of *Methanobrevibacter* (Rep2_Bin008). The lack of genome recovery of the most abundant methanogen in Rep1 may be attributed to the low abundance of methanogens (< 5%, generally) in digester communities (De Vrieze et al., 2016; Sundberg et al., 2013), as well as strain heterogeneity among methanogens (Charuvaka and Rangwala, 2011). Thus, reference genomes of *Methanosarcina* sp. (NC_003901) and *Methanocorpusculum* sp. (NR_075452) were selected as methanogenic representatives in Rep1 and Rep3, respectively. The genome of *Methanosarcina* (Rep1) harbored hydrogenotrophic, acetoclastic, and methylotrophic pathways. The *Methanobacterium* genome (Rep1), beside hydrogenotrophic pathway, also encoded acetyl-CoA synthetase (*acs*) and acetyl-CoA decarbonylase/synthase (*ACDS*) essential for the acetoclastic pathway. However, all known *Methanobacterium* species are physiologically recognized as hydrogenotrophic methanogen (Boone, 2001), where enzyme ACDS may function in reverse to carry out acetyl-CoA synthesis from CO and methyl (Adam et al., 2018). Moreover, the genera *Methanobrevibacter* (Rep2) and *Methanocorpusculum* (Rep3) genomes only encoded full hydrogenotrophic metabolism. In addition, genes for formate dehydrogenase (*fdh*) were identified in almost all genomes except in the *Methanosarcina* genome, indicating their ability to alternatively use formate as initial substrate and subsequently enter into hydrogenotrphic pathway. Collectively, the hydrogenotrophic pathway was observed in all analyzed methanogen genomes.

3.5. ROS detoxification in Proteiniphilum and methanogens

Since ROS-detoxification mechanism supported the adaptation of the putative *Proteiniphilum* strains and methanogens under microaerobic conditions, the occurrence of antioxidant genes in the genomes was also examined in this study (Figs. 1 and 2, and Table S4). All the analyzed *Proteiniphilum* genomes encoded superoxide dismutase (SOD) detoxifying the $O_2^{\cdot -}$, and catalases (CAT), glutathione peroxidases (GPX) and peroxiredoxins (PRX) reducing the produced H_2O_2 (Table S2), which belong to energy-free ROS scavenger but allow new O_2 production during their detoxification reaction (Martins et al., 2019). For methanogens, beside SOD, and CAT, GPX and PRX, energy-dependent $O_2^{\cdot -}$ scavenger (superoxide reductase, SOR) and energy-dependent H_2O_2 scavenger (rubrerythrin, Rbr) were identified to be more ubiquitously present in their genomes (Table S2). In contrast, energy-dependent ROS-quenching by SOR and Rbr enable an advantage of no oxygen formation for anaerobes, yet require additional energy sources with F_{420} as the ultimate electron donor in methanogens (Martins et al., 2019). Thus, the predominant bacterial genus *Proteiniphilum* and methanogenic archaea in the ORP-controlled microaeration AD system possessed a ROS-scavenging mechanism, thereby providing a selective advantage under microaerobic conditions.

3.6. Mitigation of VFAs accumulation by Proteiniphilum and methanogens in intermittent microaerobic digestion

Based on the genome-scale metabolic network reconstruction, we consistently observed complement sets of cellulose- and hemicellulose- hydrolyzing enzymes and facultative metabolic pathways in *Proteiniphilum* genomes, as well as prevalence of the hydrogenotrophic pathway in taxonomically diverse methanogens

in replicate samples. Consequently, cooperative interaction between *Proteiniphilum* strains and methanogens is proposed as follows (Fig. 2): 1) *Proteiniphilum* strains were involved in the breakdown of cellulose and hemicellulose, overcoming the rate-limiting hydrolysis step; 2) During intermittent microaerobic condition (every other day), when no oxygen was supplied, VFAs were produced by facultative *Proteiniphilum* strains via anaerobic fermentation. When oxygen was introduced, the VFAs were rapidly consumed by the same facultative *Proteiniphilum* strains via aerobic respiration using cytochrome-*bd* oxidase. This reduces the stress on the rate-limiting syntrophic reaction, associated with VFA-degrading syntrophs and H_2-consuming methanogens; 3) Methane production can be achieved by taxonomically diverse methanogens; 4) ROS (e.g., $O_2^{\cdot-}$ and H_2O_2) scavengers are ubiquitously identified in both *Proteiniphilum* strains and methanogens, releasing the oxidative stress during microaerobic conditions.

4. Discussion

Our previous study demonstrated stable AD operation at high OLRs using the strategy of intermittent (every other day) microaeration (picomolar oxygen) regulated by the ORP level of +25 mV above anaerobic baseline (Nguyen et al., 2019). Within a 24-h microaerobic cycle, an ORP level of -470 mV, -450 mV and -475 mV was maintained in three independent replicates, respectively. In other words, repeated start-up/shut-down cycles of oxygen injection were executed, and an approximate DO level of $1.98^{*}10^{-8}$ mg/L (or 0.62 pM) was maintained during the microaerobic cycle (Nguyen et al., 2019). We evaluated why the only dominant bacteria were *Proteiniphilum* strains, but a diverse range of methanogens were able todominate under intermittent exposure of picomolar oxygen in replicate experiments.

4.1. Oxygen resistance of cytochrome bd-encoded Proteiniphilum and diverse methanogens

Proteiniphilum strains encode solely one terminal oxidase, cytochrome *bd* oxidase, for direct oxygen consumption and energy conservation. Generally, cytochrome *bd* is noteworthy for its high affinity to oxygen among different aerobic terminal oxidases (D'mello and Poole, 1996). Baughn and Malamy (2004) demonstrated that the cytochrome *bd* in the nanaerobe *Bacteroides fragilis,* was essential for its oxygen-dependent growth at a concentration as low as 300 nM, but inhibited its viability at concentrations higher than 2 μM. Furthermore, due to the up-expression of cytochrome *bd* oxidase under microaerobic conditions (Tseng et al., 1996), *Escherichia coli* was found to fuel the aerobic respiration at oxygen concentrations lower than 3 nM (measurement based on precision sensor). In addition, an oxygen concentration as low as 0.05 nM was explored based on an oxygen flux modelling calculation (Stolper et al., 2010). These findings are in agreements with our observation that cytochrome *bd*-encoded *Proteiniphilum* strains highly enriched at the extreme microaerobic condition of 0.62 pM. More importantly, this study proposes the occurrence of aerobic respiration at an oxygen level of one to two orders of magnitude lower than previously reported, which shows much broader environmental range of aerobic organisms in the ecosystems. However, a time-delayed gene regulatory network for fermentative and respiratory switch was observed in a cytochrome *bd*-encoded *E. coli* K-12 during the initial phase of transition from anaerobic to microaerobic conditions (Trotter et al., 2011). In the study, the introduction of oxygen resulted in an immediate inhibition of FNR-controlled PFL (anaerobic pyruvate fermentation), but a 60-120 mins delay in induction of PdhR-regulated PDHC activity (aerobic pyruvate oxidation) (Trotter et al., 2011). Hence, it suggests that suitable oxygen exposure pattern (i.e., 24-hr intermittent microaerobic operation) is pivotal in alternative expression of metabolic switch between fermentation and aerobic respiration in *Proteiniphilum*. In addition to direct oxygen consumption, antioxidant enzymes-encoding genes (e.g., *SOD, CAT, GPX, PRX*) were also ubiquitous in *Proteiniphilum* genomes, implying that it has the ability to further scavenge the concurrent oxygen-induced ROSs under microaerobic conditions. Overall, the metabolism of cytochrome *bd*-dependent aerobic respiration and oxygen detoxification contributed together to the oxygen tolerance of bacterium *Proteiniphilum* in intermittent (every other day) oxygen-limited (picomolar level) system. Based on these finds, we hypothesize that the consistently predominant facultative bacterium *Proteiniphilum* readily performs aerobic respiration during the oxygen supply period, thereby releasing the stress of VFAs accumulation, which relies solely on syntrophs in traditional AD, by converting a portion of VFAs to CO_2.

For methanogens, though an aerobic respiration pathway was not found, various antioxidant enzymes for both $O_2^{\cdot-}$ and subsequent H_2O_2 detoxification were detected in the taxonomically diverse methanogens. Considering their close interaction with the facultative oxygen scavenger *Proteiniphilum* it may rapidly consume the majority of the dissolved oxygen, shielding the surrounding methanogenic cells from direct oxygen attack, but releasing a small amount of ROSs during oxidative phosphorylation process for further toxic effects. Therefore, this may induce the occurrence of antioxidant enzymes in the methanogens. More intriguingly, energy-dependent $O_2^{\cdot-}$ and H_2O_2 reductase (F_{420} as ultimate electron donor), SORs, and Rbrs, were more ubiquitously present in these methanogens than energy-free ROS scavengers (SOD, CAT, GPX and PRX). Although such energy-dependent ROS scavengers confer a selective advantage (no new O_2 production during detoxification reaction) on anaerobes, they need to divert the electron flow from their energy source in order to reduce oxidative stress (Jenney Jr. et al., 1999). Thorgersen et al. (2012) reported that anaerobic archaea *Pyrococcus furiosus* used electrons from sugar oxidation to reduce O_2 via SOR, resulting in a decrease in fermentative H_2 production. On this basis, electrons from methanogenesis that are normally used to generate CH_4 were likely diverted to detoxify ROS in the studied methanogens. Moreover, with various redox potential sets in different replicates (-470 mV, -450 mV and -475 mV, methanogens might utilize different amounts of diverted energy to relieve the oxidative toxicity at different levels, therefore requiring diverse methanogens to adapt to their ORP-specific cellular energy status. Actually, antioxidant systems are critically important for poising the extra- and intracellular redox milieus, which are maintained far from equilibrium and reflect electron transfer and redox balance between cell membrane (Banerjee, 2012). And, a shift in environmental redox potential can impact the distribution of endogenous and exogenous ROS, as well as cellular redox status, thereby regulating the associated metabolism (Adler et al., 1999). Hence, the application of various sets of ORP values in Reps 1, 2, and 3 (-470 mV, -450 mV and -475 mV) might have led to the dominance of different methanogenic taxa (*Methanosarcina, Methanobrevibacter*, and *Methanocorpusculum*, respectively) in this study. Buan (2018) reported that methanogens required a reducing environment with a redox potential below + 50 mV. However, the methanogenic species-specific adaptation to various ORP levels needs to be studied further.

4.2. Microbial survival niches under intermittent oxygen-poor environments

Fig. 3 illustrates an intermicrobial relationship in the intermittent microaerobic AD process. And, such microbial prevalence and interaction may also be shaped by the species-specific growth

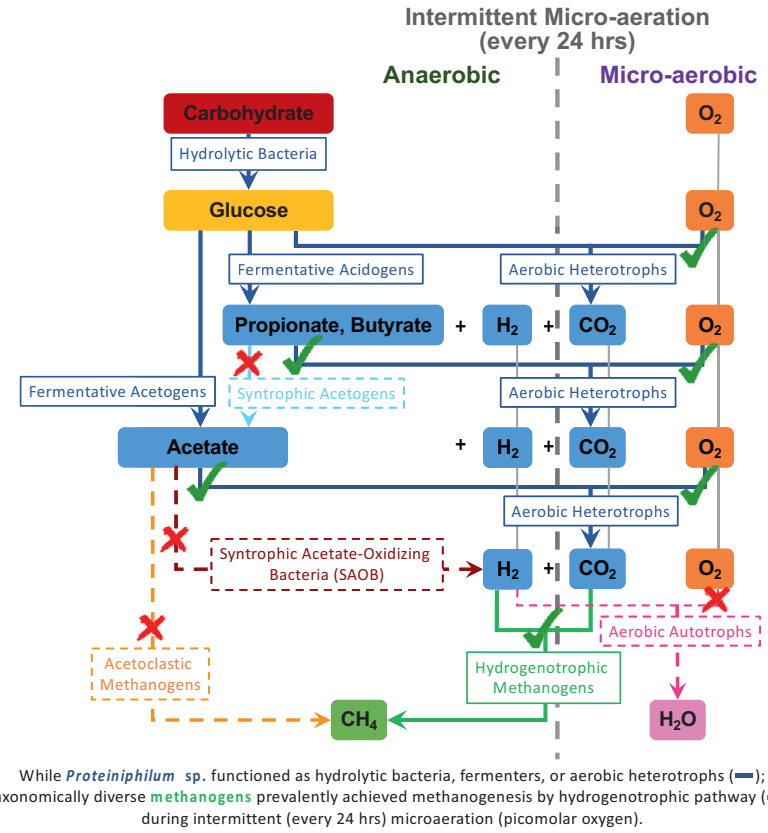


Fig. 3. Illustration of interspecies interactions within microaerobic digester community. Biomethane production from lignocellulosic biomass, which was mainly accomplished by the synergistic action of *Proteiniphilum* (dark blue solid lines) and hydrogenotrophic methanogen (green solid line), outcompeting syntrophic acetogens (light blue dashed line), syntrophic acetate-oxidizing bacterium (SAOB, brown dashed line), acetoclastic methanogens (orange dashed line) and aerobic H_2-oxidizing autotrophs (pink dashed line) in their growth kinetics and thermodynamics, when intermittent microaeration was applied. The bacterium *Proteiniphilum* (dark blue solid lines) has the ability to function as a hydrolytic bacterium, a fermenter, or an aerobic heterotroph under either strictly anaerobic or microaerobic conditions.

properties. On the one hand, facultative heterotroph like *Proteiniphilum* have both kinetic (μ_m = 13.2 *vs.* ~ 0.35 day^{-1}) and thermodynamic ($\Delta G'_R$ = ~ -90.0 *vs.* > 0.0 kJ/e^-) advantages over syntrophic fermenters (i.e., syntrophic propionate- and acetate-oxidizing bacteria) for degrading accumulated VFAs (Lawrence and McCarty, 1969; Nguyen et al., 2019; Rittmann and McCarty, 2001). On the other hand, even though the facultative heterotrophic *Proteiniphilum* and hydrogen-oxidizing bacteria are at the same level in energy acquisition ($\Delta G'_R$ = ~ -90.0 *vs.* -92.7 kJ/e^-), the former grows much faster than the latter (μ_m = 13.2 *vs.* 10.1 day^{-1}) (Matassa et al., 2015; Nguyen et al., 2019; Rittmann and McCarty, 2001), thereby making them more competitive in oxygen scarce environments. Here, the insufficient oxygen supply during microaerobic condition is reflected by the fact that VFAs reduced but remained in the system. Hence, the operational mode of picomolar oxygen injection by intermittently elevating ORP of +25 mV above anaerobic baseline contributed to the predominance of facultative heterotroph *Proteiniphilum* in the system.

In addition, the niche difference between acetoclastic and hydrogenotrophic methanogens in ORP controlled microaeration ecosystem is also of great interest. In this study, we observed that hydrogenotrophic pathway, rather than acetoclastic one, is more prevalent in the methanogens studied (Fig. 1), implying the dominance of the hydrogenotrophic pathway. Here, functional redundancy but taxonomic diversity was observed in the methanogenic archaeal communities. The thermodynamic fundamentals underlying such hydrogenotrophic dominance could be similar to studies by Dolfing et al. (2008) and Mayumi et al. (2013), in which increased CO_2 partial pressure makes hydrogenotrophic pathway much more exergonic over acetoclastic one. Nevertheless, contrary results on methanogenesis were found between our and their studies. While we proposed hydrogenotrophic dominance, they reported that the acetoclastic methanogens instead of syntrophic acetate-oxidizing bacteria coupled with hydrogenotrophic methanogens dominated in their CO_2-injected oil reservoir environment based on stable isotope-based metabolic profiling and community analysis (Dolfing et al., 2008; Mayumi et al., 2013). The discrepancy could be attributed to the fact that whether syntrophic acetate-oxidizing bacteria (SAOB) participated in the two scenarios (CO_2 injection *vs* microaeration). Direct CO_2 injection drives more endergonic syntrophic acetate oxidation, making it a rate-limiting step for methanogenesis via hydrogenotrophic pathway (Mayumi et al., 2013). Meanwhile, microaeration facilitates the exergonic aerobic acetate oxidation outcompeting the higher thermodynamic constraint of SAOB, and promotes the dominance of hydrogenotrophic methanogenesis in this study. Low abundance of SAOB in our microbial community analysis supports the above illustration. However, further studies using a stable isotope-based metabolic profiling as well as metatranscriptomic or metaproteomic analysis are needed to further explore these possibilities.

4. 3. Engineering implications

The underlying microbial mechanisms elucidated in this study support our earlier findings that, intermittent (every other day) microaeration (picomolar oxygen) has significant potential to maintain digester stability (neutral pH level with no alkalinity addition) at high OLRs in field applications. This study demonstrates

that a picomolar concentration of oxygen maybe sufficient to introduce the aerobic respiration in AD to rapidly scavenge accumulated VFAs. Thus, microaeration could be applied to other complex substrates and co-digestion systems to enhance digester stability and to achieve higher methane yields. Although mixtures of oxygen and methane (at oxygen concentrations of 5-15% by volume) may have some safety concern, the oxygen in gas phase of ORP-based microaeration system was often much less than 1%. Furthermore, when adequate and standard safety measures are incorporated into plant design, any significant threat from an explosion hazard can be minimized. Moreover, microaeration may only be initiated when the digester shows signs of instability at high OLR. Once the digester is brought back to stability, microaeration could be ceased.

5. Conclusions

This study revealed a robust analysis of metabolic networks between *Proteiniphilum*-affiliated bacteria and methanogenic archaea under picomolar oxygen conditions by intermittently setting ORP at +25 mV above anaerobic baseline, an approach that enabled efficient digestion of lignocellulosic biomass at high OLR. The cytochrome *bd* encoded in the nanaerobe *Proteiniphilum* may periodically perform aerobic respiration to degrade the VFAs accumulated during anaerobic fermentation so as to maintain stability at high OLRs. Furthermore, the ubiquitous presence of antioxidant enzymes in *Proteiniphilum*-affiliated bacteria and different methanogenic taxa may have contributed to detoxification of oxygen for stable methane yield. However, to verify the functional activities between swift aerobic-anaerobic switch, further investigations are necessary and would need to employ tools such as stable isotope probing (SIP), metatranscriptomic or metaproteomic approaches. In summary, the metabolic pathway proposed here advances our insights in the microbial interactions in the intermittent microaerobic digestion and facilitates process design and operational strategies when developing a high rate AD system.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2020.116721.

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